

1) One-way ANOVA

Consider an agricultural experiment where treatments are different levels of water availability for plants

Crop yield is the response

$$Y_{ij} = \mu_i + \varepsilon_{ij} \quad \begin{matrix} j = 1, \dots, n_i \\ i = 1, \dots, r \end{matrix}$$

$r = \# \text{ treatments}$
 $n_i = \# \text{ replicates}$ } randomized experiment

Assumption: $\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$

Question 1: Is there a treatment effect?

$H_0: \mu_1 = \dots = \mu_r$ H_A : not all μ_i 's are equal

Test for presence of treatment effect:

$$F = \frac{MSTR}{MSE}$$

If $\mu_i = \mu_0 + \alpha_i$
 with $\sum n_i \alpha_i = 0$
 then $\hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}_{..}$

$$\text{where } SSTO = SSTR + SSE$$

$$\sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2 \quad \sum_i n_i (\underbrace{\bar{Y}_{i.} - \bar{Y}_{..}}_{\alpha_i})^2 \quad \sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2$$

$$df(SSTO) = \sum n_i - 1$$

$$df(SSTR) = r - 1$$

$$df(SSE) = \sum n_i - r$$

Beyond asking whether there is a treatment effect, we may want to

- (i) infer on the average yield corresponding to particular treatment.
- (ii) compare average yields of pairs of treatments.
- (iii) infer on contrasts of average yields corresponding to different treatments.

Let $\mu = (\mu_1, \dots, \mu_r)$ be a parameter vector

A (linear) contrast involving the vector μ is an expression of the form $\sum_{i=1}^r c_i \mu_i$ where $c_1, \dots, c_r$ are specific numbers with $\sum_{i=1}^r c_i = 0$

eg. $\mu_1 - \mu_2 \checkmark$ $\mu_1 - 2\mu_2 + \mu_3 \checkmark$
 $\mu_1 - \mu_2 + \mu_3 \times$ $= (\mu_1 - \mu_2) - (\mu_2 - \mu_3)$

Inference on treatment means:

(a) $100(1-\eta)\%$ confidence interval for μ_i

$\hat{\mu}_i = \bar{Y}_i$ Fact: $\bar{Y}_i - \mu_i \sim N(0, \frac{\sigma^2}{n_i})$

So, $\frac{\bar{Y}_i - \mu_i}{\sqrt{MSE/n_i}} = \frac{\bar{Y}_i - \mu_i}{\sqrt{\frac{\sum_{j=1}^r \chi_{\sum n_j - r}^2}{\sum n_j - r}}} \sim t_{(\sum_{j=1}^r n_j - r)}$ Let $N = \sum_{i=1}^r n_i$

Then $100(1-\eta)\%$ CI for μ_i is given by

$$\bar{Y}_{i.} \pm \sqrt{\frac{MSE}{n_i}} t_{N-r}^{1-\eta/2}$$

(b) Find C.I. for the treatment effect α_i

BLUE for α_i is $\hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}_{..}$

$$\begin{aligned} \text{Var}(\hat{\alpha}_i) &= \text{Var}(\bar{Y}_{i.} - \bar{Y}_{..}) \\ &= \text{Var}\bar{Y}_{i.} + \text{Var}\bar{Y}_{..} - 2\text{Cov}(\bar{Y}_{i.}, \bar{Y}_{..}) \\ &= \frac{\sigma^2}{n_i} + \frac{\sigma^2}{N} - \frac{2\sigma^2}{N} \left\{ \begin{aligned} &\text{Cov}(\bar{Y}_{i.}, \bar{Y}_{..}) \\ &= \text{Cov}\left(\bar{Y}_{i.}, \frac{\sum n_i \bar{Y}_{i.}}{N}\right) \\ &= \text{Cov}\left(\bar{Y}_{i.}, \frac{n_i \bar{Y}_{i.}}{N}\right) \\ &= \frac{n_i}{N} \text{Var}\bar{Y}_{i.} \\ &= \frac{n_i}{N} \frac{\sigma^2}{n_i} = \frac{\sigma^2}{N} \end{aligned} \right. \\ &= \frac{\sigma^2}{n_i} - \frac{\sigma^2}{N} \end{aligned}$$

Thus $100(1-\eta)\%$ CI for α_i is
 $(\bar{Y}_{i.} - \bar{Y}_{..}) \pm \sqrt{MSE} \sqrt{\frac{1}{n_i} - \frac{1}{N}} t_{N-1}^{1-\eta/2}$

(c) Find C.I. for a contrast $\sum c_i \mu_i = \theta$

BLUE for θ is $\hat{\theta} = \sum c_i \hat{\mu}_i = \sum c_i \bar{Y}_{i.}$

$$\hat{\theta} - \theta \sim N\left(0, \left(\sum_{i=1}^r \frac{c_i^2}{n_i}\right) \sigma^2\right)$$

So $100(1-\eta)\%$ CI for θ is given by

$$\sum c_i \bar{Y}_{i.} \pm \sqrt{MSE} \sqrt{\sum \frac{c_i^2}{n_i}} t_{N-r}^{1-\eta/2}$$

Multiple Comparison/Simultaneous Inference :

E.g. we may be interested in all, or a subset of the parameters $\mu_i - \mu_j$ for $i < j$

(total of $\binom{r}{2}$ parameters)

Constructing confidence set for individual pairwise difference, or equivalently, performing hypothesis test of individual parameters raises the question of whether the confidence level is maintained simultaneously.

Let $\theta_1, \dots, \theta_g$ be the individual parameters of interest

Let $C_{1-\eta_k}(\underline{Y})$ be a $100(1-\eta_k)\%$ C.I. for θ_k for data

$k=1, \dots, g$

This means that

$$P_{\mu, \sigma^2}(\theta_k \in C_{1-\eta_k}(\underline{Y})) \geq 1-\eta_k \quad \text{for } k=1, \dots, g$$

We want $100(1-\eta)\%$ simultaneous ①

confidence set for $(\theta_1, \dots, \theta_g)$ i.e., we want a set say $C_{1-\eta}(\underline{Y}) \subset \mathbb{R}^g$ s.t.

$$P_{\mu, \sigma^2}((\theta_1, \dots, \theta_g) \in C_{1-\eta}(\underline{Y})) \geq 1-\eta \quad \text{--- ②}$$

If we restrict to rectangular confidence sets then

$C_{1-\eta}(\underline{Y})$ is of the form

$$C_{1-\eta_1}(\underline{Y}) \times C_{1-\eta_2}(\underline{Y}) \times \dots \times C_{1-\eta_g}(\underline{Y})$$

i.e., eqnⁿ (2) becomes

$$\begin{aligned} P_{\mu, \sigma^2}((\theta_1, \dots, \theta_g) \in \bigotimes_{j=1}^g C_{1-\eta_j}(\underline{Y})) &\geq 1-\eta \\ &= P_{\mu, \sigma^2}\left(\bigcap_{j=1}^g \{\theta_j \in C_{1-\eta_j}(\underline{Y})\}\right) \end{aligned}$$

Question, what values of $\eta_1, \dots, \eta_g$ will ensure that

$$\begin{aligned} P\left(\underbrace{\bigcap_{j=1}^g A_j}_{\substack{j=1 \\ \dots \\ g}}\right) &\geq 1-\eta \quad \text{if } P(A_j) \geq 1-\eta_j \\ &\quad \substack{j=1, \dots, g} \\ &= 1 - P\left(\bigcup_{j=1}^g A_j^c\right) \\ &\geq 1 - \sum_{j=1}^g P(A_j^c) \\ &\geq 1 - \sum_{j=1}^g \eta_j \quad \text{by (1)} \end{aligned}$$

So by Bonferroni's Inequality, if we choose $\eta_1, \dots, \eta_g > 0$ s.t. $\sum_{j=1}^g \eta_j = \eta$, we accomplish the goal.

In practice, we set $\eta_j = \eta/g$ for $j=1, \dots, g$

The resulting simultaneous inference procedure is called Bonferroni's procedure

The level $(1-\eta)$ is referred to as a family wise confidence level.